

THE HAMER LOOP & XENO-ANTIGENIC SEQUESTRATION

A Systems-Level Hypothesis for the Dietary Origins of Acne, Rosacea, and Co-Presenting Systemic Inflammation

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Abstract

This paper presents a systems-level hypothesis derived from over four decades of continuous longitudinal observation, pattern recognition, and repeated self-provocation and elimination trials. The central premise is that acne, rosacea, and a constellation of co-presenting systemic conditions — including peripheral oedema, soft tissue inflammation, nocturnal nasal congestion, obstructive sleep apnoea, and gastrointestinal disturbance — may represent predictable, externally triggered immune responses sharing a common dietary origin, rather than discrete and unrelated disorders.

The proposed mechanism centres on dietary incorporation of N-glycolylneuraminic acid (Neu5Gc), a non-human sialic acid abundant in bovine-derived foods, into host tissue glycans, where it is recognised as foreign by the immune system. This antigen-antibody interaction initiates what this paper terms the Hamer Loop: a self-reinforcing inflammatory cycle involving lymphatic sequestration, interstitial accumulation, and site-specific amplification via tissue-resident memory T cells (TRM cells). Reaction severity is further modulated by ammonium-containing food additives, which act as co-factors that lower the inflammatory threshold and amplify lesion severity.

Complete elimination of bovine-derived dietary triggers produced full resolution of the entire symptom cluster within 7–10 days across repeated trials, with marked re-exposure sensitivity upon reintroduction. The cutaneous manifestations — acne and rosacea — are proposed as the most visible presenting features of a systemic process, functioning as a biological sentinel signal rather than primary dermatological disorders. This framework is offered as a structured, observation-driven hypothesis warranting formal investigation, with direct translational application in the Acnerase dietary screening platform.

Keywords: *Neu5Gc; xenosialitis; acne; rosacea; glycomics; tissue-resident memory T cells; lymphatic stasis; ammonium potentiation; diet-immune interaction; systems biology*

1. Clinical Context & Observational Foundation

Unlike conventional research models built primarily on cohort studies and laboratory isolation, this work originates from continuous bedside assessment and real-time physiological tracking across thousands of exposure-response cycles spanning more than four decades (1981–2026). The observational record underpinning this hypothesis is therefore unusually extensive for a single-author, N-of-1 framework, and its breadth constitutes both its primary strength and its primary limitation, as discussed in Section 8.

The central premise is that acne and rosacea may, in certain individuals, represent predictable, externally triggered immune responses rather than purely localised dermatological disorders. This reframing has direct implications for diagnosis, management, and future research into chronic inflammatory disease and is developed mechanistically in the sections that follow.

1.1 The 2017 Cardiac Event: A Natural Experiment in Perfusion and Lesion Distribution

A significant systemic event in 2017 provides an instructive natural experiment within this longitudinal record. Following multiple myocardial infarctions, six coronary stents were placed at the proximal segment of the left anterior descending (LAD) artery, at the point of approximately 95% occlusion. The immediate restoration of cardiac output produced a dramatic, system-wide improvement in perfusion that had been chronically compromised, likely for years prior to the acute event.

In the nine years following stent insertion, a gradual but sustained shift in lesion distribution has been observed, with acne presentations becoming increasingly concentrated in the scalp and upper head-neck territory. Prior to 2017, lesion distribution had spanned the conventional sites of face, neck, shoulders, back, and chest across several preceding decades. This distribution shift is mechanistically consistent with the lymphatic hypothesis central to the Hamer Loop model: the scalp maintains an exceptionally rich lymphatic network draining to the cervical and occipital nodes, and restored cardiac output — by increasing arterial pulsatility and interstitial fluid turnover — would be expected to re-activate lymphatic cycling preferentially in highly vascularised territories.

This observation cannot be controlled for all confounding variables, consistent with the N-of-1 limitations acknowledged in Section 8. It is documented as a plausible, mechanistically coherent marker within an otherwise continuous longitudinal record, constituting a natural experiment embedded within the observational dataset.

2. The Biological Mismatch: Neu5Gc and Human Immunity

Humans uniquely lack the functional CMP-N-acetylneuraminic acid hydroxylase (CMAH) enzyme, and therefore cannot endogenously synthesise N-glycolylneuraminic acid (Neu5Gc) — a sialic acid abundant in most non-human mammals. Neu5Gc can nonetheless be introduced exogenously through dietary sources, principally:

- Red meat (bovine sources being the primary trigger in this framework)
- Dairy products and dairy derivatives
- Bovine gelatin and other bovine-derived food additives

Existing glycobiology research, notably by Varki and colleagues at the University of California San Diego, has demonstrated that dietary Neu5Gc can be metabolically incorporated into human tissue glycans, where it is recognised as foreign by the immune system (Dhar et al., 2019; Tangvoranuntakul et al., 2003). This creates conditions for chronic, low-grade antigen-antibody interaction, forming the proposed biological basis for what this paper terms Xenosialitis: a persistent immune response to incorporated non-human sialic acids.

While Neu5Gc is present in most mammals, longitudinal self-provocation trials (1981–2026) have identified bovine-derived sources as the primary inflammatory triggers in this subject, while porcine, ovine (lamb), and caprine (goat) sources remained non-reactive. The mechanistic basis for this bovine specificity is addressed in Section 2.2.

2.1 The Evolutionary Origin of the CMAH Mutation

The human loss of CMAH function is not recent or random. Genomic evidence places the inactivating mutation approximately 2–3 million years ago, after the divergence of the human lineage from our last common ancestor with chimpanzees. Every other extant great ape — chimpanzees, gorillas, bonobos, and orangutans — retains functional CMAH and produces Neu5Gc endogenously. Humans are the sole exception.

The leading hypothesis, advanced by Varki and colleagues, proposes that the selective pressure was pathogen-driven. Specifically, an ancestral relative of *Plasmodium reichenowi* — a malaria-causing pathogen that infects chimpanzees — is believed to have used Neu5Gc as a molecular binding site to gain entry into red blood cells. Early humans who carried the CMAH-inactivating mutation lacked surface Neu5Gc on their erythrocytes, rendering them resistant to this pathogen. In an environment where the infection was lethal, the survival advantage conferred by CMAH loss was sufficient to drive the mutation to near-universal prevalence across the human population (Chou et al., 1998; Varki, 2009).

The evolutionary irony embedded in this account is directly relevant to the Hamer Loop framework. The very mutation that protected early humans from a lethal haemoprotozoan infection created a new biological condition: a species that cannot synthesise Neu5Gc, has no immune tolerance for it, and yet consumes it in large quantities through the diet. The anti-Neu5Gc antibodies that drive Xenosialitis are not a pathological aberration — they are the expected immunological consequence of a body encountering a molecule it has been genetically programmed to treat as foreign for approximately two to three million years.

2.2 Bovine Specificity Within the Neu5Gc Response

Not all dietary Neu5Gc sources produce equivalent inflammatory responses in susceptible individuals, and the reason lies in molecular context rather than Neu5Gc presence alone. Bovine tissue contains Neu5Gc at unusually high concentrations relative to porcine, ovine, and caprine sources, and presents it within glycan architectures that appear to be particularly immunogenic in humans. The specific configuration of bovine glycan structures — the molecular scaffolding surrounding the Neu5Gc molecule — determines how efficiently circulating anti-Neu5Gc antibodies bind and initiate the inflammatory cascade.

This bovine specificity is further supported by an observation involving farmed Atlantic salmon, which is commercially fed rations containing rendered bovine products including blood-based pellets. Despite the bovine-derived content of the feed, consumption of farmed salmon did not trigger an inflammatory response across longitudinal self-provocation trials. The proposed explanation is that secondary metabolic processing by a non-mammalian host sufficiently alters the bovine glycan structures such that the resulting Neu5Gc is no longer presented in the configuration recognised by the human immune response. This observation, taken together with the porcine and ovine negative controls, points toward a specific molecular signature of bovine Neu5Gc — rather than Neu5Gc broadly — as the operative trigger in this framework.

3. The Hamer Loop: A Systems-Level Model

3.1 Reframing the Problem

Conventional dermatology attributes acne and rosacea primarily to sebum overproduction, follicular occlusion, and bacterial colonisation. While these factors account for downstream pathology, they do not adequately explain:

- Rapid lesion onset following exposure to specific foods
- Reproducible and consistent timing patterns across separate exposure events
- Lesion recurrence at identical anatomical sites across separate episodes
- Co-presenting multi-system symptoms involving nasal, gastrointestinal, musculoskeletal, and neurological domains

The Hamer Loop proposes that these conditions are instead externally triggered, immune-mediated inflammatory cycles, with the skin serving as a distal expression site of a systemic process. In this model, the cutaneous manifestation is not the disease — it is the signal. The skin functions as a biological canary: the most visible indicator of a systemic antigenic burden whose consequences extend well beyond the dermis.

3.2 Reproducible Clinical Timeline

Across repeated exposures spanning four decades, a consistent temporal sequence emerges. This pattern has remained stable across elimination and re-exposure trials, suggesting a structured physiological mechanism rather than non-specific reactivity.

Stage	Timing	Observed Response	Proposed Mechanism
Stage 1	Minutes	Rhinorrhoea, flushing, vasomotor activation	Mast cell degranulation; histamine release; capillary permeability ↑
Stage 2	~20 minutes	Abdominal bloating, gastrointestinal discomfort	Systemic cytokine distribution; early interstitial fluid shift
Stage 3	2–4 hours	Lesion formation at consistent anatomical sites	TRM cell activation; lymphatic sequestration; interstitial reservoir formation
Stage 4	Overnight	Lesion amplification (facial predominance)	Recumbent fluid redistribution; increased interstitial pressure; nocturnal amplification

Table 1. Reproducible Clinical Timeline Following Dietary Neu5Gc Exposure. This sequence has remained stable across repeated provocation trials spanning 1981–2026.

3.3 Proposed Mechanistic Sequence

3.3.1 Xeno-Antigen Exposure and Incorporation

Dietary Neu5Gc enters systemic circulation following intestinal absorption and is metabolically incorporated into glycan structures on endothelial and epithelial cells. This creates a hybrid molecular signature in which host tissue presents a non-human antigen — the foundational condition for Xenosialitis.

3.3.2 Immune Recognition and Engagement

Circulating anti-Neu5Gc antibodies bind to incorporated Neu5Gc in host tissues, generating immune complexes that activate complement and Fc receptor-bearing effector cells. This interaction produces a diffuse, low-grade inflammatory state, consistent with the observed systemic symptoms that precede and accompany cutaneous involvement.

3.3.3 Mast Cell Activation and Vascular Response

Early-stage symptoms — rhinorrhoea, flushing, and vasomotor activation within minutes of trigger ingestion — suggest rapid mast cell degranulation and histamine release. Histamine increases vascular permeability, allowing inflammatory mediators to shift from the intravascular compartment into interstitial spaces. This vascular opening is the gateway to the sequestration events that follow.

3.3.4 Interstitial Sequestration and Lymphatic Dynamics

Inflammatory molecules accumulate in localised interstitial regions as lymphatic clearance capacity is overwhelmed. Regional lymphatic flow becomes functionally impaired, inflammatory compounds are not efficiently cleared, and a localised inflammatory reservoir forms. Rather than

resolving, this reservoir persists and amplifies — the defining characteristic of the Hamer Loop as a self-reinforcing cycle.

3.3.5 Tissue-Resident Memory T Cells and Site Recurrence

One of the most reproducible features of this model is lesion recurrence at identical anatomical coordinates across separate exposure events. This aligns precisely with the established behaviour of tissue-resident memory T cells (TRM cells), which persist in previously inflamed tissue, respond rapidly upon re-exposure to their cognate antigen, and direct immune activity to specific anatomical sites. Within the Hamer Loop, TRM cells function as localised amplifiers — providing the mechanistic explanation for site-specific recurrence that conventional dermatological models do not address.

4. The Ammonium Potentiation Effect

A key differentiating feature of this model is the observation that reaction severity is not solely dependent on Neu5Gc exposure. Repeated provocation trials have consistently demonstrated that ammonium-containing compounds — commonly present in processed foods as flavour enhancers, colour stabilisers, and processing aids — act as co-factors that amplify the primary immune response in ways that Neu5Gc load alone does not predict.

4.1 Observed Effects

- Lowered threshold for triggering an inflammatory response: exposures that would remain subclinical in the absence of ammonium become clinically apparent
- Disproportionate severity: reactions to small exposures in ammonium-containing products exceed what the Neu5Gc load alone would predict
- Production of deep nodular lesions: the highest-severity presentations in the observational record are consistently associated with co-exposure to both Neu5Gc sources and ammonium-derived additives

4.2 Proposed Mechanisms

The following mechanisms are proposed on the basis of mechanistic plausibility and observational consistency, pending formal biochemical validation:

- Alkalinisation of mucosal pH, disrupting the acid-barrier defence against antigen penetration
- Direct mast cell sensitisation, lowering the degranulation threshold for Neu5Gc-antibody interactions
- NLRP3 inflammasome priming, amplifying IL-1 β and IL-18 output upon antigen presentation
- Enhanced antigen presentation efficiency, rendering incorporated Neu5Gc more accessible to circulating anti-Neu5Gc antibodies

The Ammonium Potentiation Effect is documented in detail in Document 2 of this series, including the lesion morphology hierarchy associated with different trigger combinations. The Acnerase ingredient database identifies over 150 ingredient synonyms and processing terms associated with ammonium exposure, many of which are not apparent from label reading alone.

5. Nocturnal Amplification

Lesion severity consistently increases overnight following an active Hamer Loop cycle. This nocturnal amplification is proposed to arise from three intersecting physiological mechanisms:

- Recumbent fluid redistribution: in the supine position, interstitial fluid that accumulated in the face, neck, and chest during the active cycle redistributes toward these territories under hydrostatic pressure
- Increased interstitial pressure: the redistributed fluid increases pressure within already-inflamed tissue, mechanically amplifying the inflammatory reservoir
- Pilosebaceous unit stress: prolonged decubitus creates mechanical pressure within follicular structures, contributing to lesion progression at Stage 3 sites

This nocturnal pattern aligns with the clinically observed ‘morning flare’ phenomenon — the consistent observation that lesion severity and surface oiliness are maximal upon waking — and supports a hydrostatic contribution to lesion development that is distinct from but additive to the immunological mechanisms described above.

6. Sentinel Observations

6.1 The Gelatin-Antihistamine Paradox

A particularly instructive observation emerged during a period of antihistamine use for symptom management. A gelatin-encapsulated diphenhydramine formulation (Benadryl) worsened rather than attenuated the inflammatory response. Substitution with an identical dose in caplet form — without a bovine-gelatin capsule shell — produced the expected antihistamine effect and lesion resolution within 12–24 hours.

The mechanism is direct: bovine gelatin is a concentrated source of Neu5Gc. Administration of a gelatin-encapsulated antihistamine during an active Hamer Loop flare introduces a further antigenic bolus at the precise moment the immune system is being asked to de-escalate. This paradoxical aggravation — when systematically documented — constitutes a mechanistically coherent natural experiment. It also has direct clinical implications: susceptible individuals should be aware that standard antihistamine capsule formulations may contain the antigen the antihistamine is intended to address.

6.2 Negative Controls and Antigen Specificity

To isolate the causative agent, longitudinal self-provocation trials included the ingestion of non-bovine mammalian and marine sources across several decades. The following were identified as consistent negative controls, producing no observable inflammatory cutaneous response:

- Ovine (lamb) and caprine (goat): regular consumption over four decades, zero lesion formation
- Porcine (pig): ingestion of pork products and porcine-derived gelatin, no Hamer Loop sequence observed
- Marine (fish): fish-derived proteins and gelatins, non-reactive
- Farmed salmon (secondary bovine exposure): no inflammatory response despite documented bovine-derived feed — consistent with secondary metabolic processing neutralising the bovine antigen

These negative controls establish that the immune response in this subject is specifically directed at bovine-derived Neu5Gc in its unmodified configuration, not at Neu5Gc broadly, mammalian-derived food generally, or animal fat as a class.

6.3 The Olfactory Inversion

A striking and previously undocumented observation concerns the reversal of olfactory valence following complete trigger clearance. During periods of high Neu5Gc load, bovine-derived food stimuli — specifically baked goods containing cheese or dairy fats in a bakery environment — were perceived as intensely pleasurable and appetising. Following complete systemic purging of the trigger load over 7–10 days, the identical olfactory stimulus was perceived as noxious and aversive, prompting immediate avoidance rather than attraction.

This olfactory inversion — from craving to aversion in response to the same stimulus, triggered solely by antigen clearance — may reflect the unmasking of the immune system's aversive signalling once chronic antigen habituation is removed. It is proposed to explain, in part, why dietary elimination is psychologically difficult for susceptible individuals: the triggering foods are actively craved during the chronically loaded state. To the author's knowledge, this phenomenon has not been previously documented in the context of dietary Neu5Gc exposure and warrants formal investigation.

6.4 The Systemic Resolution Profile

During deliberate elimination trials — undertaken as part of a modified Atkins dietary protocol in which all primary triggers (wheat, beef, and dairy derivatives) were simultaneously removed — the following systemic resolution was observed within 7–10 days. This profile is documented here in full because its multi-system simultaneity is itself the most powerful argument for a single upstream mechanism.

System	Observation
Cutaneous	Complete resolution of acne and rosacea lesions; complete resolution of oily skin

System	Observation
Lymphatic / vascular	Resolution of peripheral oedema (hands, fingers, anterior tibial region); resolution of calf tightness
Musculoskeletal	Complete resolution of joint and body pain previously attributed to arthritis
Gastrointestinal	Resolution of abdominal bloating and loose stools
Respiratory / sleep	Resolution of nocturnal nasal congestion, snoring, and obstructive sleep apnoea
Neurological / sensory	Olfactory inversion as described in Section 6.3

Table 2. Systemic Resolution Profile Following Complete Dietary Trigger Elimination. All systems resolved within the same 7–10 day window, consistent with a single upstream mechanism.

The clinical significance of the musculoskeletal finding warrants specific note. What had been attributed to degenerative joint disease resolved completely on dietary elimination, without pharmacological intervention, within the same timeframe as the cutaneous and gastrointestinal symptoms. Resolution on elimination and return on re-exposure is a strong indicator of inflammatory — not degenerative — aetiology. The arthritis misattribution may represent one of the more clinically consequential downstream effects of unrecognised xenosialitis.

The sleep apnoea finding is similarly notable. All alcoholic beverages examined contain dairy or wheat derivatives as colour enhancers, flavour enhancers, sulphites, or processing agents. Resolution of sleep apnoea on alcohol elimination is therefore consistent with removal of the Neu5Gc/ammonium trigger load rather than alcohol-specific mechanisms — a connection not currently recognised in sleep medicine literature and representing a testable hypothesis.

6.5 Re-Exposure Sensitivity: The Empty Tank Phenomenon

Once systemic trigger load was fully cleared — approximately 7–10 days — re-exposure to even a small quantity of trigger foods produced a markedly amplified response, disproportionate to the dose. The author describes this as analogous to a fuel tank: chronically full of triggers, the body adapts to a persistent baseline load and the inflammatory signal is partially masked. Once purged, the threshold drops dramatically and even trace exposure produces a strong response.

This phenomenon has direct implications for elimination trial protocols and for patient counselling. The commonly observed ‘tolerance’ to these foods in regular consumers may represent physiological adaptation to chronic load — masking the underlying inflammatory process rather than reflecting genuine non-reactivity. Individuals who have successfully completed an elimination protocol should be advised that whole-food Neu5Gc sources, which produce only attenuated responses in the chronically loaded state, may trigger a pronounced acute response after clearance.

7. Translational Application: The Acnerase Platform

The Acnerase platform translates this hypothesis into a practical preventive tool, available at no cost as an Android application. Its core function is real-time ingredient identification: by scanning a product's label, the application identifies Neu5Gc-containing ingredients, ammonium-derived catalysts, and wheat-derived inflammatory proteins (gliadins and ATIs), and classifies the product's inflammatory potential using the Neu5Gc Severity Index documented in Appendix II of this series.

7.1 Key Design Principles

- On-device processing (edge computing architecture): no data transmitted externally
- Zero data collection from users
- No cost, no advertising, no external funding
- Proprietary database of 150+ ingredient synonyms covering Neu5Gc sources, ammonium compounds, and gluten/ATI indicators

The platform's goal is prevention through informed dietary avoidance, rather than treatment of established lesions. A separate toggle identifies gliadins and ATIs for users managing wheat sensitivity or celiac-related conditions, including dermatitis herpetiformis.

8. Limitations

The primary limitation of this work is its N-of-1 observational basis. While the longitudinal depth of the dataset is substantial — spanning 45 years, thousands of exposure-response cycles, and deliberate controlled provocation trials — the absence of a controlled cohort means that individual confounding variables, including genetic susceptibility, microbiome variation, and concurrent dietary exposures, cannot be fully excluded. Self-provocation trials, though systematic, are inherently subject to reporting bias.

The Ammonium Potentiation Effect and the nocturnal amplification mechanism are, at this stage, exploratory. They are included because of their predictive consistency across observations, but each requires independent validation before mechanistic claims can be formalised.

The bovine specificity finding — while supported by decades of negative controls across porcine, ovine, caprine, and marine sources — is derived from a single subject and cannot be generalised to the susceptible population without cohort data.

This paper does not present clinical proof. It presents a structured, reproducible observational framework intended to generate testable hypotheses for formal investigation. All mechanistic claims are offered as hypotheses, not established facts, and are hedged accordingly throughout this document.

9. Conclusion

The Hamer Loop does not seek to replace established dermatological models, but to address gaps that they do not fully account for. It proposes that acne and rosacea may, in some individuals, represent repeatable, diet-triggered, immune-mediated inflammatory cycles with systemic origins — and that the skin, in these individuals, is functioning not as the primary site of disease but as its most visible signal.

The multi-system resolution profile documented in Section 6.4 — simultaneous resolution of acne, rosacea, joint pain, peripheral oedema, sleep apnoea, and gastrointestinal disturbance on a single dietary elimination protocol — is not the profile of unrelated conditions resolving by coincidence. It is the fingerprint of a single upstream mechanism reasserting homeostatic control.

The hypothesis is immediately testable at the individual level: remove suspected triggers, observe symptom resolution, and reintroduce to monitor response. This creates an accessible bridge between complex molecular theory and real-time personal validation — and it is precisely this bridge that the Acnerase platform is designed to provide.

Position Statement: *“Neu5Gc exposure and immune interaction likely occur broadly across the population. Clinical expression appears to depend on individual susceptibility, threshold effects, and co-factors. In susceptible individuals, this may present as visible inflammatory skin manifestations; in others, it may remain subclinical or contribute to background inflammatory burden. The skin is the canary. The mine is the body.”* — Martin Hamer, RN (Ret.)

10. Proposed Validation Framework

The N-of-1 observational basis of this framework, while unusually extensive in its longitudinal depth, remains the primary limitation of the evidence presented. The following six investigations are proposed to elevate the Hamer Loop hypothesis from structured observation to formally testable science. Each maps directly to a specific mechanistic claim and each is falsifiable: a negative result would constitute meaningful evidence against the relevant component of the hypothesis.

10.1 Serum Mast Cell Tryptase During Active Flare

Serum mast cell tryptase is the gold-standard biomarker for systemic mast cell degranulation. Elevated tryptase during a controlled provocation trial — drawn at the Stage 1–2 window within two hours of trigger ingestion — with return to baseline between flares, would directly confirm the mast cell component of the model and formally distinguish the Hamer Loop from chronic baseline mast cell activation disorders.

10.2 Serum Complement Levels (C3, C4) During and Between Flares

If anti-Neu5Gc antibodies are binding to incorporated Neu5Gc and forming immune complexes, complement will be consumed in the process. Depressed C3 and C4 with elevated split

products C3d and C4d during active flares, returning to normal between them, is the biochemical fingerprint of Type III hypersensitivity — one of the most diagnostically clean confirmatory findings available in clinical immunology. A paired within-subject design measuring complement during a controlled provocation and again at baseline following a 14-day elimination would provide direct mechanistic evidence for immune complex formation as the initiating event.

10.3 Anti-Neu5Gc Antibody Titres

Validated ELISA assays for serum anti-Neu5Gc IgG have been developed by Varki and colleagues at UC San Diego. The key experiment is not merely confirming anti-Neu5Gc antibody presence — these exist in most humans — but demonstrating that titres are meaningfully elevated in acne and rosacea patients relative to matched controls, and that titres correlate with flare severity and dietary Neu5Gc exposure. A titre spike following controlled provocation, with progressive decline during a subsequent elimination period, would constitute the most powerful individual serological finding available to this hypothesis.

10.4 Skin Biopsy at 2–4 Hours: Immune Complex and Mast Cell Staining

Standard dermatological biopsies are taken from established lesions and stained for bacteria — an approach that reliably confirms the conventional model by design. The Hamer Loop framework requires a fundamentally different protocol: tissue taken from an early-phase lesion at 2–4 hours post-provocation, processed for direct immunofluorescence staining for IgG, IgA, and C3 deposition, and stained for mast cell tryptase rather than bacterial culture. Immune complexes present in the dermis at this early stage — before conventional bacterial colonisation has established — would directly challenge the primacy of the bacterial model and provide anatomical confirmation of the immune complex hypothesis.

10.5 Urinary Neu5Gc Quantification

Dietary Neu5Gc can be detected in urine following consumption of bovine-derived foods, providing a non-invasive measure of metabolic incorporation and excretion. Measuring urinary Neu5Gc before, during, and after a controlled provocation trial — in subjects with acne or rosacea versus matched controls without inflammatory skin conditions — would address whether metabolic incorporation differs between susceptible and non-susceptible individuals, and whether urinary Neu5Gc excretion correlates with flare onset and resolution.

10.6 Near-Infrared Fluorescence Lymphatic Imaging During Active Flare

Near-infrared fluorescence (NIRF) lymphatic imaging permits non-invasive, real-time visualisation of superficial lymphatic flow in human subjects. Applied during an active Hamer Loop flare and compared to a cleared baseline state, NIRF imaging could directly visualise the regional lymphatic stasis proposed as the sequestration mechanism in Section 3.3.4. This investigation has, to the author's knowledge, never been attempted in the context of acne or rosacea research and represents a genuinely novel experimental application of an established imaging modality.

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